

Comparison of Wiener Filtering and Spectral Subtraction for Speech Noise Reduction

Student	Zhizhong Chen	Student ID	530668932
GitHub username	czz0806	Course	ELEC5305
Repository	https://github.com/czz0806/elec5305-project-530668932		
Project site	https://czz0806.github.io/elec5305-project-530668932/		

Approximate body word count: 804 (excluding student information, timeline and references).

1. Project Overview

Background noise reduces speech clarity, listening comfort and the reliability of downstream speech systems. This project proposes a controlled comparison of two established single-channel speech-enhancement techniques: Wiener filtering and spectral subtraction. Real clean speech and recorded environmental noise will be selected from the Microsoft Deep Noise Suppression (DNS) Challenge resources rather than generated with a text-to-speech system or recorded specifically for this project. The signals will be mixed at several known input signal-to-noise ratios (SNRs), processed using both algorithms in MATLAB, and evaluated under identical conditions. The central question is: how do Wiener filtering and spectral subtraction differ in noise reduction, speech distortion and perceptual quality when the noise type and input SNR change?

2. Background and Motivation

Speech enhancement aims to estimate a clean speech signal from a recording corrupted by additive noise. Spectral subtraction is a historically important approach because it is transparent and computationally light. It estimates a noise spectrum during non-speech or noise-only frames and subtracts that estimate from the magnitude or power spectrum of the noisy signal. Boll (1979) demonstrated the practicality of this method, but subtraction errors can leave isolated spectral components that are perceived as musical noise. A spectral floor can reduce this problem, although stronger suppression may increase speech distortion.

Wiener filtering instead applies a frequency-dependent gain derived from estimates of speech and noise power. Under the assumed statistical model, the gain minimises mean-square estimation error. Its performance therefore depends heavily on reliable noise and a priori SNR estimation (Scalart and Filho, 1996). These different mechanisms make the two methods suitable for a focused comparison that demonstrates short-time Fourier analysis, noise estimation, spectral modification and reconstruction without requiring a large neural network. The DNS Challenge is appropriate because it provides openly documented clean-speech and real-noise resources designed for noise-suppression research (Reddy et al., 2021).

3. Proposed Methodology

3.1 Data source and preparation

A manageable subset of approximately eight clean utterances from multiple speakers and several recorded noise types will be selected from the DNS Challenge resources. Only the files required for the experiments will be downloaded; the full corpus will not be duplicated in the GitHub repository. Source links, licences and selection criteria will be documented. Audio will be converted to mono, resampled to 16 kHz where necessary, trimmed to comparable durations and peak-normalised. Each clean utterance will be mixed with a noise recording at planned input SNRs of -5, 0, 5 and 10 dB. Using known mixing levels preserves a clean reference for objective evaluation.

3.2 Signal-processing implementation

Both enhancement methods will use the same short-time Fourier transform (STFT) front end so that the comparison is fair. An initial configuration will use 32 ms Hamming-windowed frames with 50% overlap, followed by inverse STFT and overlap-add reconstruction. For spectral subtraction, the noise spectrum will be estimated from known noise-only material, subtracted from each noisy-speech spectrum, and limited by a small spectral floor to avoid negative magnitudes. For Wiener filtering, speech and noise power estimates will be used to form a gain of the form $G(k,m) = P_s(k,m) / [P_s(k,m) + P_n(k,m)]$. The gain will be applied to the noisy complex spectrum before reconstruction. Parameters such as the subtraction factor, spectral floor and noise-smoothing rate will be recorded rather than tuned separately for favourable examples.

3.3 Evaluation and reproducibility

For every method, noise type and input SNR, the output will be compared with the clean reference. Primary quantitative measures will be output SNR and SNR improvement; segmental SNR and PESQ will be included if the required MATLAB functions or authorised implementations are available. Waveforms and spectrograms will show residual noise and speech-component loss, while short listening checks will document audible distortion and musical noise. Results will be summarised in consistent tables and plots. MATLAB scripts will be separated into data preparation, enhancement, evaluation and plotting modules, with fixed random seeds and configuration values to support reproducibility.

4. Expected Outcomes

The expected deliverables are a reproducible MATLAB prototype for both algorithms, selected audio examples, before-and-after spectrograms, quantitative comparison tables and a documented GitHub project site. The project does not assume in advance that one algorithm will always be superior. Wiener filtering is expected to provide smoother attenuation when its power estimates are accurate, whereas spectral subtraction is expected to be simpler but more vulnerable to musical-noise artefacts. The experiments should show how these trade-offs change with SNR and real noise type. Success will mean that another student can follow the documentation, obtain the authorised source data, run the scripts and reproduce the comparison.

5. Feasibility and Risk Management

The scope is feasible within one semester because it compares two classical methods on a deliberately small data subset and does not require model training. The main risks are excessive dataset size, inconsistent sampling formats, weak noise estimates and unavailable perceptual-metric software. These will be managed by downloading only selected files, using one preprocessing pipeline, retaining a spectral floor, and treating PESQ as an additional metric rather than the sole success criterion.

6. Timeline

Weeks	Planned activity
1-2	Confirm topic, research question and project scope.
3-5	Complete literature review; select and document DNS Challenge audio samples.
6-7	Implement preprocessing and spectral subtraction; verify reconstruction.
8-9	Implement Wiener filtering; establish common experimental settings.
10-11	Run SNR/noise experiments; calculate metrics; create plots and listening examples.
12-13	Interpret results; finalise report, code documentation and GitHub project site.

7. References

1. Boll, S. F. (1979). Suppression of acoustic noise in speech using spectral subtraction. *IEEE Transactions on Acoustics, Speech, and Signal Processing*, 27(2), 113-120. <https://doi.org/10.1109/TASSP.1979.1163209>
2. Loizou, P. C. (2013). *Speech Enhancement: Theory and Practice* (2nd ed.). CRC Press.
3. Reddy, C. K. A., Dubey, H., Koishida, K., Nair, A., Gopal, V., Cutler, R., Braun, S., Gamper, H., Aichner, R., & Srinivasan, S. (2021). INTERSPEECH 2021 Deep Noise Suppression Challenge. *Proceedings of Interspeech 2021*, 2796-2800. <https://doi.org/10.21437/Interspeech.2021-1609>
4. Scalart, P., & Filho, J. V. (1996). Speech enhancement based on a priori signal-to-noise estimation. *Proceedings of the IEEE International Conference on Acoustics, Speech, and Signal Processing*, 2, 629-632. <https://doi.org/10.1109/ICASSP.1996.543199>

8. Proposed Processing Flow

Stage	Planned output
1. Data selection	Authorised clean-speech and real-noise samples documented from DNS Challenge resources.
2. Preprocessing	Mono, 16 kHz, duration-matched and normalised source signals.
3. Controlled mixing	Noisy test signals at -5, 0, 5 and 10 dB input SNR.
4. Enhancement	Spectral-subtraction and Wiener-filter outputs using common STFT settings.
5. Evaluation	SNR results, waveforms, spectrograms, listening notes and reproducible MATLAB scripts.